Uncover the Scoring Secrets of the “Black Pearl Restaurant Guide” – Use Mathematical and Statistical Methods to Analyze

By Xiuning Gu

1. Introduction

The “China Black Pearl Restaurant Guide” a representative of food guide in China, has been published seven times since its inception in 2018. The “Black Pearl Restaurant Guide”, launched by the Meituan company, is not only recognized by some food lovers, chefs, and gourmets, but also has accumulated a growing reputation among the public. Apps such as Meituan and Dianping play a crucial role in the daily lives of Chinese residents, becoming some of the most active daily users of life-service apps.

The guide is known for its rigorous criteria, judging restaurants based on multiple dimensions such as culinary skills, restaurant standards, and heritage and innovation. However, different views have emerged in the selection of the guidelines. Some people insist that not all the restaurants listed in the guide qualify as “Black Pearls” for their delicacies, while others argue that many restaurants with unique flavors are overlooked. This divergence makes it necessary to dig deeper into the mechanisms underlying the selection process.

This report aims to provide some comprehensive estimates of the weight of the three factors: culinary skills, restaurant standards, and heritage and innovation, by applying mathematical and statistical methods. Moreover, in addition to being constrained by these obvious criteria, the choice of restaurants is also influenced by other external, implicit factors, such as geographical elements and related economic factors. This provides us with a broad research area for in-depth analysis of restaurants in different regions and food categories in China, so as to more fully understand the principles behind the selection of the Black Pearl Guide.

In the following discussion, there will be some analysis methods such as linear regression as the important tool to study various variables including ratings and awards of all Black Pearl restaurants in China. Through this method, it is able to reveal the correlation between the selection results and these factors, and further analyze the scientific nature of the selection criteria. In addition, this report will compare the differences between Black Pearl Restaurants and combine various factors to attempt to build a model to predict the diamond rating.

Finally, drawing on Brent Furdyk's report on “How to become a 1, 2, or 3 Michelin star restaurants”, the “Black Pearl” will be the focus of our attention. What exactly determines whether a restaurant qualifies as a “black pearl” restaurant? This puzzle will be gradually revealed in our research, providing valuable experience and insights for those who dream of owning a “Black Pearl” restaurant. This project will uncover the mysteries.

2. Analysis methods

2.1 Introduction to Regression Analysis

Regression analysis is mainly about determining how changes in the independent variables correlate with changes in the dependent variable. [1]

Regression analysis as the primary method in my report is widely recognized as the predominant statistical technique in contemporary research. Specifically, regression analysis constitutes a substantial portion of the research methodology, with the regression model serving as its central focus.[2]

2.2 Simple Linear Regression

2.2.1 Definition and Concept:

The fundamental concept underlying regression analysis is the ability to represent the relationship between two variables, y and x, through a model. So, the case of one explanatory variable is called simple linear regression.[3] [4]

2.2.2 Mathematical Foundation:

Given the numerous variables and factors influencing the performance of the Black Pearl Restaurants, the multiple linear regression approach will be utilized to account for these complexities. The basic thought of regression model is integrated in the publicity.

$$Y = f(x) + e\,.$$

However, in the ideal case, $f(x)$ is the essential relationship between y and x, and

$$e = Y - f(x)\,.$$

e is the model error, which is represented by a constant. In generally speaking, $f(x)$ represents the part that could be explained and calculated by the regression model, that is, the direct relationship between y and x. Similarly, the linear regression model can be expressed as

$$y = a + bx + \varepsilon.$$

In this equation, "y" represents the dependent variable that we want to predict or explain, "x" is the independent variable used for prediction, "a" is the y-intercept (the value of y when x is 0), "b" is the slope of the regression line and "ε" accounts for the error term, capturing the variability in y that is not accounted for by x.[5]

2.2.3 Assumptions:

-Linearity: some form of regularization is typically necessary to prevent unreasonable solutions from arising in the estimation process. The mean of the response variable is a linear

combination of the regression coefficients and the predictor variables. [6]

-Independence of variables: The wrong part will be discarded because each result is independent, this report is looking for the relationship between them.

-Homoscedasticity: The variance or standard deviation of the forecast should be proportional to the mean of the x variable.

-random errors: "e"cannot be explained because there are three reasons respectively. Generally speaking, there are three kinds of errors. They are respectively 1. model is not good enough. 2.the model ignore other variables. 3. several random errors exists. [7]

2.2.4 Example:

My height is 182 cm, and my foot length is 28cm. So based the foot height is the independent variable, and the body height is the dependent variable, I consider that $y = 6.5\,x$. However, it is not a good model, because there are some reasons above it.

2.2.5 Method:

And the fundamental method is to use the 'least square method' of polynomial to fit the specific model to forecast and evaluate the Black Pearl restaurant, and finally draw a conclusion about which is the most important factor.

When collecting the body height and foot length of some people as the data, it can convert these data into coordinates and spot them on the coordinate axis. Then people are able to draw a line, evenly distribute the data points on both sides. Also this line is called 'the best fit line'. The idea is to write the abscissa of each piece of data as X_i, so $e_i = Y_i - (a \times X_i + b)$.

Secondly, the concept of residual sum of squares is introduced,

$$RSS = \sum_{i}^{n} {e_i}^2$$

The main reason for squaring is to avoid that positive and negative errors do not cancel each other out. So, the most accurate linear relationship can be found when RSS is equal to the minimum, just as the derivative of a quadratic function is equal to zero. And above, these are the 'least square method'. And the fundamental method is to use the "least square method" of polynomial to fit the specific model to forecast and evaluate the Black Pearl restaurant, and finally draw a conclusion about which is the most important factor. [8]

2.3 Multi-linear Regression

2.3.1 Definition and Expansion from Simple Regression:

Colloquially, when a variable has many relationships with other independent variables, this process is just called "multiple linear regression" like combined so many simple linear

regressions. Multiple linear regression is able to estimate model parameters through the above and simple linear regression least square method, mainly through matrix operations to find the best fitting plane.

2.3.2 Mathematical Foundation:

The basic model for multiple linear regression is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i, \quad for\ each\ observation\ i = 1, \ldots, n.$$

And the format written as a vector looks like this.

$$(a_1 X_1 + a_2 X_2 + a_3 X_3 + a_4 X_4 + \cdots + a_n X_n) = ((a_1 + a_2 + a_3 + a_4 + \cdots + a_n) \times \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$$

2.3.3 Limitations:

-Multicollinearity: In multiple linear regression, there may be correlation between multiple independent variables. For example, in the data of Black Pearl restaurants collected by me, there is a small relationship between the two independent variables, the number of restaurant collections and the number of reviews, or even unit linear regression. If the analysis is direct, the problem of multicollinearity may occur. Therefore, if we persist in such analysis, the parameter estimation of the whole model may become unstable, but there are also some solutions, such as increasing the sample size, such as Ridge Regression or Lasso Regression to punish to improve the stability, or stepwise regression can be used to avoid some errors. I also use these methods in my research.

2.3.4 Assumptions:

-Dimensions of data analysis: Multiple linear regression can be analyzed on higher dimensional data sets and has the same characteristics as matrix. Usually, multiple linear regression is used for many high-dimensional data. At the same time, in real life, certain events or certain data are affected by multiple independent variables and multiple relationships. For example, the evaluation of the Black Pearl restaurant in my theme study will be affected by various scores, opening time, the number of comments and other factors.

Also, in the case of small effects, each kind of variable is like an individual factor that can be regarded as independent in my view.

2.4 Hypothesis testing

In addition, the report also used hypothesis testing methods from the A-level Further Mathematics book to verify the validity of research hypotheses and inferences. Hypothesis testing is a commonly used statistical inference method to evaluate the reliability of research

conclusions by analyzing sample data to determine whether the research hypothesis is valid. With the aid of hypothesis testing, we can further confirm the reliability of the results of linear regression analysis and the conclusions of the study. In summary, this study comprehensively and in-depth analyzes the market performance and popularity of Black Pearl Restaurant in 2024 and provides sufficient support and argument for the reliability of the research conclusions.[9]

2.4.1 Definition and Concept:

It is a widely held view that hypothesis testing is a method of statistical inference that has been introduced in high school. Probabilistic statements can often be derived by testing aggregate data. It is akin to determining whether a big eater can eat 10 hamburgers per meal, prompting the design of a hypothesis test experiment to solve the problem. [10]

Null Hypothesis（H_0）: Simply speaking, it is the hypothesis about the result setting of the overall parameters.

Alternative Hypothesis（H_1）: The opposite of H_0 generally indicates the presence of significant differences in results and conditions.

P-value: It is directly compared with the Significance Level.

Significance Level: (α) It is the allowed error probability.

One-tailed Test and Two-tailed Test: In short, if $H_1 \neq 0$, the hypothesis test is a two-tailed test. If $H_1 > 0$ or $H_1 < 0$, the hypothesis test is a one-tailed test.

2.4.2Mathematical Foundation:

a) Propose the null hypothesis: Design the null hypothesis H_0 and the alternative hypothesis H_1.
b) Design and conduct tests: To be precise, appropriate tests can be selected, for example, to compare whether the means of two groups of continuous variables are significantly different can be used with t-distribution. Testing the correlation between at least two variables can be done with a chi-square distribution. The F-test in ANOVA can be used to analyze the variance of two population samples.
c) It is to extract the key statistical variables of the experiment.
d) It is to calculate the corresponding probability (p-value) of the statistical variable according to the probability distribution under the null hypothesis.
e) Decision-making: If the test statistic exceeds the critical value or the p-value is less than the significance level, the null hypothesis is rejected. Otherwise, refuse to fail. [11]

I endeavored to identify the most suitable test for my research and data. To achieve this, I decided to start by narrowing down the scope of the search method, to distinguish between nonparametric test and parametric test. The main factor that determines the difference between

them is whether they make assumptions about the population distribution, namely, whether they are non-parametric. Upon analyzing the data, I found that it deviated from a normal distribution. Therefore, I conducted some exploration on the non-parametric test to better suit the characteristics of the data.

2.4.3 Methods for determining the normal distribution:

1. By drawing the data through histograms and other similar graphs, the distribution form of the data can be directly observed, from which to judge whether the data shows the trend of normal distribution.

2. 68-95-99.7 Rule: In simple terms, about 68% of the data values fall within one standard deviation of the mean plus or minus, about 95% of the data values fall within two standard deviations of the mean plus or minus, and about 99.7% of the data values fall within three standard deviations.

3. Sometimes, some experiences often use the Shapiro-Wilk test.

4. The method of statistical description could be used to directly calculate some basic statistics of data by using some tools like Excel and, such as sample mean μ, standard deviation σ, variance σ^2, skewness S, and kurtosis K, to make basic judgments
The first two are also very cost-effective.

2.5 Non-parametric test

2.5.1 definition：

It is noticeable that non-parametric often means statistical inference that makes no specific assumptions about overall distributions. At the same time, the Wilcoxon Test tests whether there is a difference between two dependent samples. In other words, it is a non-parametric statistical test used to compare two independent groups. [12]

“Non-parametric”denotes statistical inference methods that do not make specific assumptions about the form of the population distribution. For non-parametric tests, after understanding and exploring the Wilcoxon Rank Sum Test, Wilcoxon Signed Test, and the Wilcoxon Signed-Rank Test for Paired Samples, which are three subtly different test methods, will be applied in my study to analyze the relationship between the number of Black Pearl restaurant awards and the number of annual star-rated merchants.

First of all, it is generally likely that there is a link between the two awards, the so-called correlated sample. While the Wilcoxon Rank Sum Test is not applicable because it deals with two completely independent samples, among collected Black Pearl restaurant data, there is a very strong correlation between the number of the awards of Black Pearl restaurants and the number of annual starred merchants. The Wilcoxon Signed Test is also excluded because it only uses a single sample to determine whether there is a significant difference in the median. The Wilcoxon Signed-Rank Test becomes the winner.

2.5.2 Wilcoxon Signed-Rank Test for Paired Samples:

The paired-sample Wilcoxon signed-rank test is a statistical method used to compare whether the medians of two related samples are significantly different. The two paired observations are calculated and sorted according to the difference, and the test determines the distribution of these ranks and compares them. As in the previous description, the obtained p-value from this test is also compared to the significance level, typically set at 0.05, to determine whether the median difference between paired samples is statistically significant. If the p-value is less than the significance level, it indicates that the median of the paired samples is significantly different. On the other hand, if the p-value is greater than the significance level, it indicates that there is not enough evidence to conclude that the median difference is significant. [13]

3. Data Collection: how to quantify your factors.

3.1 Collection Process

To be precise, the Black Pearl restaurants list was recently released on the January 25th, 2024, and before that, the data backup can only be used to sort out the list of Black Pearl restaurants released in 2023. An observing of the previous years' list revealed minimal changes, with the addition of numerous new Black Pearl restaurants to the current list being the primary noteworthy change. At the same time, on many websites like “World Bank”, no historical data for Black Pearl is available, let alone the latest data. Only some Michelin-related data can be found to provide some similar "peers" help for future reference, so the data of this paper has to be collected slowly by hand. We spent 6 hours sorting out data from the Black Pearl homepage of the Meituan app. These data contain detailed information such as region, city, cuisine classification, Black Pearl Diamond level, total score, three ratings for taste, environment and service, the number of collections of the restaurant on the app, the number of reviews, the opening time, the number of times it has won the “Star Merchant of the Year”, the number of times it has included the “Black Pearl Award”, and the “per capita price” that everyone is concerned about, and the number of branches of that restaurant in a specific city.

These data can be directly collected on the Meituan app, which can be basically defined as fundamental information that conforms to the original intention of Meituan, It aims to provide a publicity platform for merchants and facilitate customers to find, compare, select and evaluate stores. There is a major reason why these data evaluations are truly filled out by merchants and customers, contributing to a large sample size. Meanwhile, Meituan has its own self-developed algorithm to calculate very balanced data, which is relatively objective. The platform's spread, influence and authenticity are highly recognized by both customers and merchants, making the “Black Pearl Restaurants Guide” the direction of their efforts in recent years. However, in addition to these basic data, they only give three simple sentences for more data behind this list, their selection process and some considerations: Cooking production (food material quality, cooking skills, five senses presentation, catering collocation, production stability), service environment (service level, environmental facilities, service stability), inheritance and

innovation (Inheritance is not old-fashioned, innovation is not forgotten; System complete personal style). For this kind of "trade secret", the so-called "general data" and linear regression could certainly help me to solve this question.
After six hours of collection, the data for 180 stores was put into Excel. Nearly half of the data was still lying scattered in the Meituan app. My tutor recommended me to write some code in Python to automatically collect the data. But resistant app, with a lot of encryptions, posed a challenge as the data was scattered on different web pages, ultimately resulting in failure.

Hence, the majority of the final data has been painstakingly curated by hand, making it essentially first-hand data. After all, there is no existing compilation by predecessors, and it can be seen as a collection specifically tailored to some of the top-notch restaurants in China. It not only serves as a crucial part of data analysis but also stands as a meaningful and insightful food guide. The upcoming analysis is expected to unveil more profound insights and conclusions from the data that cannot be gleaned simply by browsing through the Meituan app.

Another piece of first-hand data comes from a questionnaire I designed for "Opinions about Black Pearl Restaurant". There are some basic screening questions in the first place, such as "Do you know the Black Pearl restaurant guide?" and "Have you tried Black Pearl restaurant?", followed by questions such as "Where did you try it?" and "How many times have you tried it?". Relatively private questions such as "monthly income", "Do you think the pricing of the Black Pearl restaurant is reasonable?", "What are the scores of the three rating criteria for the Black Pearl restaurant?", and "Opinions on the Black Pearl list" were asked.

These data from the questionnaire will provide me with some real ideas of the public about the "Black Pearl Restaurants". It is likely to build a foundation to support some opinions and conclusions from the Excel "Black Pearl Restaurants data". Also, it represents a part of the public's thoughts as a sample survey.

3.2 Approach about how to quantify?

Firstly, it is noticeable that everyone's taste cannot be quantified or judged because of different strokes for different folks. For instance, someone who prefers to eat some low-salted food will think this restaurant makes it terrible and others may hold the opposite view. However, in other views, quantifying the taste of an individual can be challenging, but the taste of a group of people can be effectively quantified.

This is why many restaurants gather feedback and ratings from customers after their meals – to effectively measure customer satisfaction. The process of quantifying this feedback involves assessing the number of positive and negative reviews, striking a balance between them, and then presenting the findings to the restaurant's management and owners. It is widely thought that this enables them to make data-driven decisions about product features and marketing strategies. In summary, although individual tastes may be difficult to quantify, aggregate preferences of groups can be effectively quantified through data analysis and research.

In a similar vein, the Meituan app utilizes its own algorithm to translate the evaluations of thousands of consumers into a 5-star rating, while still retaining the original reviews in the comments section. By quantifying data in this way, restaurants can gauge their overall satisfaction levels and implement necessary improvements. Additionally, companies can use surveys and analysis to quantify the preferences of their target market for products. Meanwhile, consumers can employ quantification and evaluation to determine which stores they want to patronize.

3.3 Process：

Among these factors, total score, taste score, environment score and service score kept unchanged. These were put into linear regression and combined with the questionnaire together.

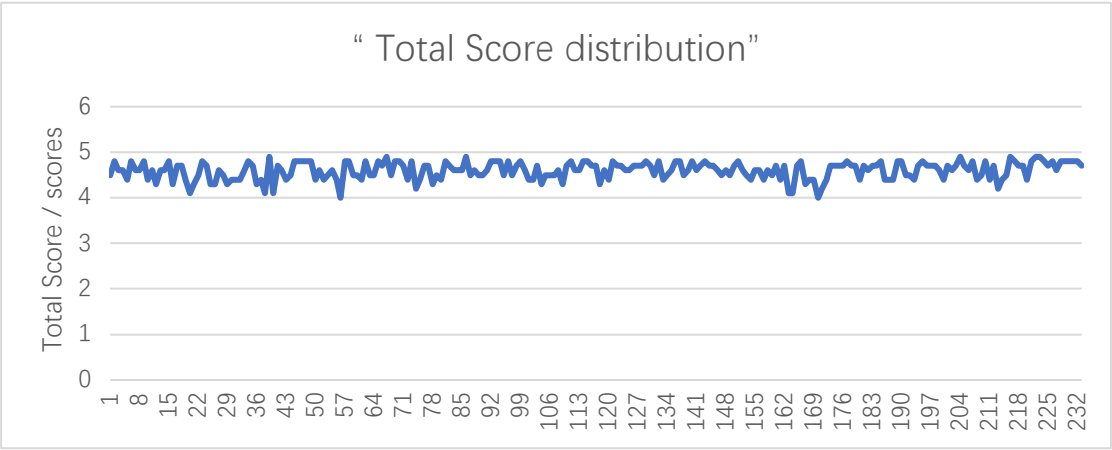
For example, # of evaluation, “Won the Annual Star Merchant ‘times’ “and “Black Pearl Restaurant List ‘times’” are temporarily put into the study according to the original data, and their parameters are step by step. # of favorites is so much bigger than others, so it was divided by 1,000 to balance with other factors due to # of favorites ranging between ten thousand and three million.

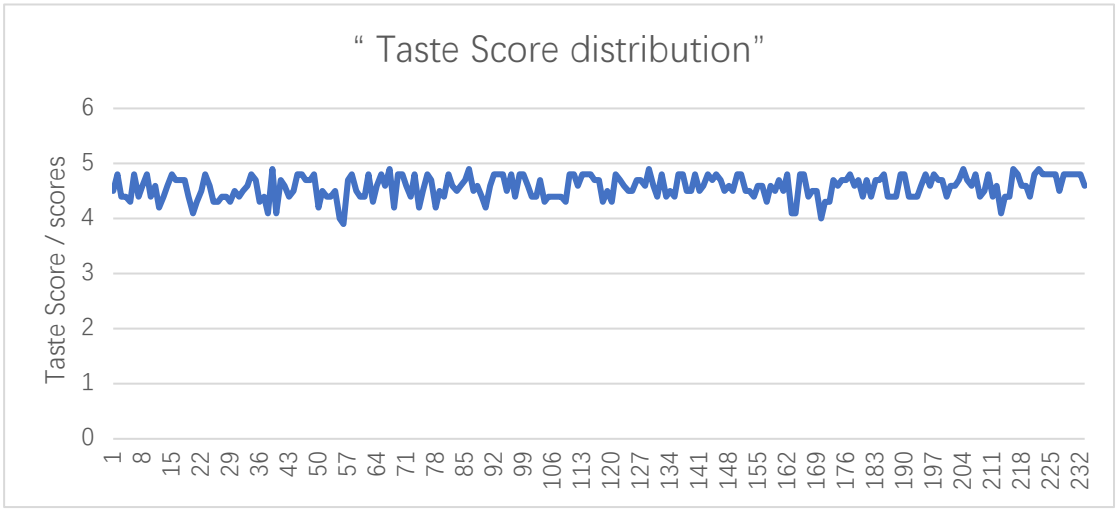
There are other factors that may not be as strongly correlated, such as of store opening time, price per capita. These were put into the hypothesis testing before testing in linear regression.

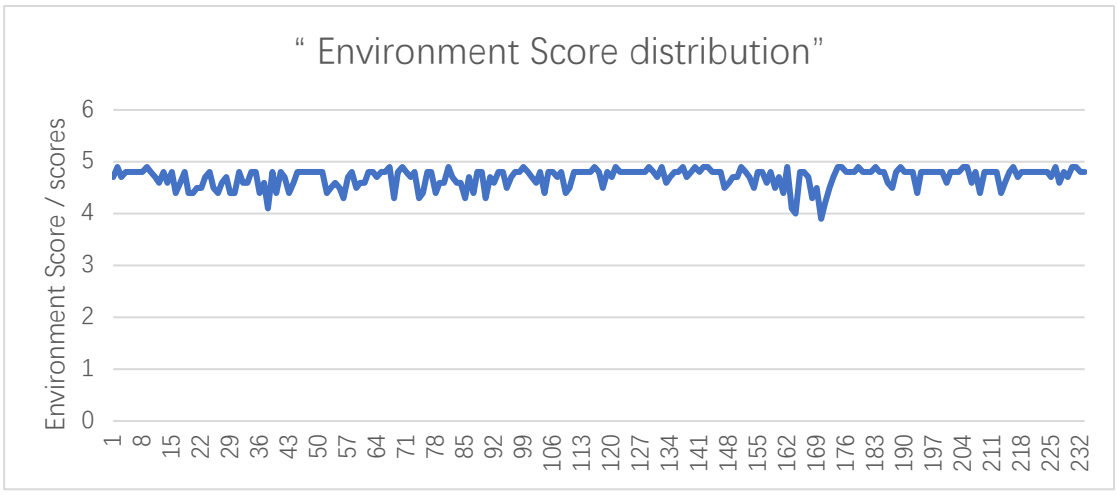
4. Implementation and analysis

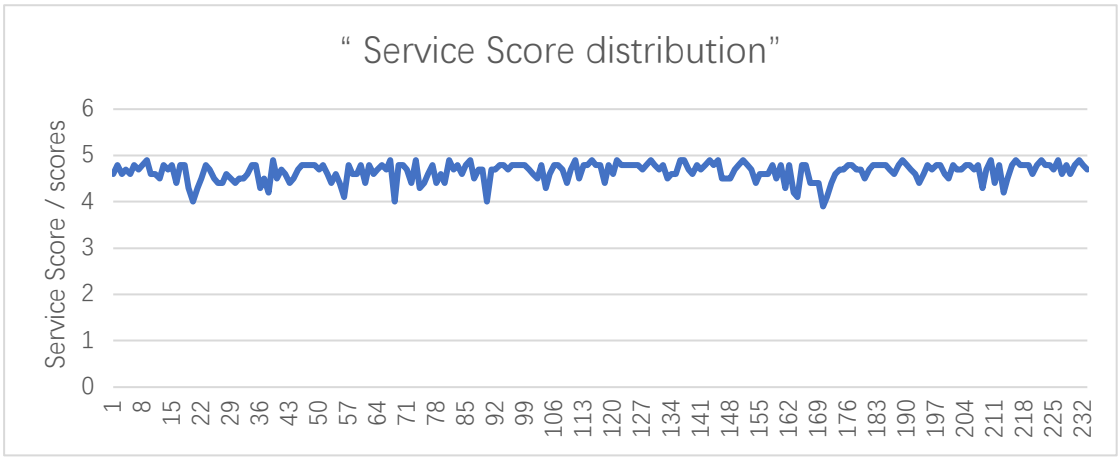
The following charts and sentences are derived from the data of “Black Pearl Restaurants” for analysis.

4.1 preliminary analysis









It is obvious that the data of each rating is maintained in the range of $[4.0, 4.9]$ points, from the four-line graphs.

	Total Score	Taste Score	Environmen	Service Score	# of favourites	# of evaluation	Time / years	Won the "annual star merchant"	"Black Pearl Restaurant " times	¥ Average price per person
mean	4.60	4.57	4.70	4.66	68134.66	3042.83	7.08	3.49	3.95	911.43
mode	4.8	4.8	4.8	26.3	26300	3	4	7	980	N/A
median	4.6	4.6	4.8	4.7	45800	1862	6	4	4	733

1. The environment receives the highest average score at 4.70, while taste and service scores are slightly lower at 4.57 and 4.66, respectively. This indicates that the Black Pearl Restaurants maintain a consistently high level of restaurant ambiance, each with its own distinctive features. However, there is some variability in taste and service scores, with overall higher recognition for the environment.

2. It is noticeable that the mode and median of the taste score are both 4.8, indicating a stable evaluation of taste, with the majority of customers providing high ratings. Moreover, most of the listed restaurants exhibit a high culinary standard. The mode of the service score is also 4.8, while the median is 4.7. Despite a 0.1 difference, the overall view suggests that the service is still commendable.

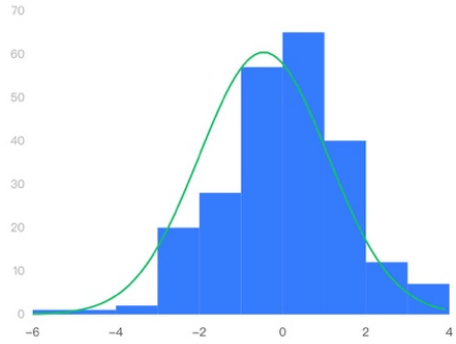
3. The average number of collections per restaurant is approximately 68,134.66, while the that of reviews is about 3,042.83. This suggests that, on average, restaurants garner considerable attention and reviews. However, some restaurants still receive lower levels of attention and reviews. Notably, the highest number of collections reached an astonishing 375,000, while the lowest was only 1,000, almost a 375-fold difference. Similarly, for reviews, one restaurant received 2,000 valid evaluations, whereas the lowest had only 100, showing a substantial difference. It is widely acknowledged that both collection and review numbers, to some extent, reflect a restaurant's popularity and perceived quality. Hence, there may be some correlation between these two factors needed to be analyzed.

The data indicates that the average operational duration of Black Pearl Restaurants is unexpectedly 7.08 years, with a median of 6 years. Most of the listed restaurants have a significant operating history, indirectly suggesting that a successful restaurant requires a considerable period of establishment. Certainly, there are also some recently opened restaurants that, like a newborn calf unafraid of a tiger, exhibit boldness in their early stages.

4.2 Hypothesis Testing:

1. The sample sizes for "Annual Star Merchant" and "Black Pearl Restaurant Guide" are less than 5000. After conducting the Shapiro-Wilk test with a significance level resulting in a P-value of 0.000***, the null hypothesis of normal distribution is rejected.

2. Given the non-normal distribution of data, the paired sample Wilcoxon signed-rank test is deemed appropriate.



	Replication of Paired Difference Normality Test Results						
Variable name	sample size	mean	std.	skewness	kurtosis	S-W test	K-S test
Won the "annual star merchant" times	233	3.494	1.495	-0.441	-0.398	-------	-------
"Black Pearl Restaurant List" times	233	3.948	2.141	0.094	-1.415	-------	-------
Won the "annual star merchant" times, "Black Pearl Restaurant List" times	233	-0.455	1.542	-0.181	0.525	0.958(0.0(	0.148(0.000***)
Note: ***, **, * represent 1 per cent, 5 per cent and 10 per cent significance levels, respectiv							

	Paired samples Wilcoxon signed rank test						
paired variable	median±std.	z	df	P	Cohen's d		
pair 1	pair 2	pairing difference (pair 1- pair 2)					
Won the "annual star merchant" times, "Black Pearl Restaurant List" times	4±1.495	4±2.141	0±1.542	4.327	232	0.000***	0.246
Note: ***, **, * represent 1 per cent, 5 per cent and 10 per cent significance levels, respectiv							

The results of the Wilcoxon signed-rank test for the two-paired sample indicate that the significance p-value for the variables “Annual Star Merchant”and “Black Pearl Restaurant List” is 0.000***, representing a significant level, thereby rejecting the null hypothesis. This illustrates a significant difference between “Annual Star Merchant” and “Black Pearl Restaurant List.” The computed Cohen's d value is 0.246, demonstrating a relatively small effect size. The awards “Annual Star Merchant” and “Black Pearl Restaurant List” share similarities, as restaurants obtaining recognition in the Black Pearl Restaurant Guide are likely to receive the “Annual Star Merchant” award as well.

4.3Chi-square test

Field name/analysis item	Phi	Crammer's V	Number of columnar links	lambda
Area	1.377	0.616	0.809	0.4

The table above presents the results of the chi-square test, including the frequency, percentage, chi-square value, and significance p-value for model validation.

1. Analyzing whether the model exhibits significance ($P < 0.05$).
2. If significance is present, rejecting the null hypothesis indicates a significant difference between the samples. Specific descriptions are made based on the percentage differences in categories. Conversely, if no significance is found, the data has no significant difference.

The results of the Pearson chi-square test analysis reveal a significance p-value of 0.000*** for the variables Cuisine and Area, demonstrating statistical significance. The null hypothesis is rejected, indicating a significant difference in the data for Cuisine and Area.

Based on the results of the effect size analysis, we can conclude that there is a strong association between Area and Cuisine. The Cramer's V coefficient is 0.616, and the phi coefficient is 1.377. For example, the only two restaurants listed in Nanchang of Jiangxi Province, both serve local "Gan" cuisine, a distinctive cuisine in Southeastern China, while only 10% of restaurants in Shanghai offer authentic Shanghainese dishes. Therefore, there is a strong overall correlation between Cuisine and Geography, but significant differences still exist.

4.4Linear regression

Linear regression analysis results n=233										
	Unstandardised coefficien	standard constant	t	P	VIF	R^2	adapt R^2	F		
B	standard error	Beta								
constant	0.182	0.109	-	1.664	0.097*	-	0.892	0.891	F=633.473	P=0.000***
Taste Score	0.617	0.032	0.676	19.56	0.000***	2.543				
Environment Score	0.196	0.04	0.189	4.887	0.000***	3.173				
Service Score	0.144	0.043	0.151	3.323	0.001***	4.385				
implicit variable: Total Score										
Note: ***, **, * represent 1 per cent, 5 per cent and 10 per cent significance levels, respectively.										

Based on the results of the linear regression analysis, it reveals a robust model. This is primarily due to the highly significant p-value of 0.000*** from the F-test, rejecting the null hypothesis of a regression coefficient being zero. It shows the model's statistical significance. Concurrently, the variables exhibit good collinearity, with all VIFs well below 10, signifying the absence of multicollinearity issues and a well-constructed model.

Analyzing the non-standardized and standardized coefficients, we observe significant impacts of Taste Score, Environment Score, and Service Score on the Total Score. Their standardized coefficients (Beta) are 0.617, 0.196, and 0.144, respectively. This implies varying degrees of influence on the Total Score, with the Taste Score having the most substantial impact. The model's R-squared is 0.891, indicating the model can explain 89.1% of the data variance, signifying a well-fitted model.

In conclusion, it can be inferred that the Total Score for Black Pearl restaurants can be predicted using the following formula:

$$\boldsymbol{Total\ Score} = \mathbf{0.182} + \mathbf{0.617} \times \boldsymbol{Taste\ Score} + \mathbf{0.196} \times \boldsymbol{Environment\ Score} + \mathbf{0.144} \times \boldsymbol{Service\ Score}$$

All these score components significantly contribute to the overall rating. Therefore, this formula serves as a predictive measure for the rating standards of Black Pearl Restaurants.

4.5Mixed Model (If divided by regions)

Comparison	Per Capita Price Difference
Shanghai vs. North China	No significant difference, small variance
Other regions vs. Shanghai	Significant differences with:
	Central (Substantial)
	Southwest, Jiangsu-Zhejiang (Moderate)
	South (Smaller)
North China, Southwest, and other region comparisc	Significant differences with:
	Central(Substantial)
	Jiangsu-Zhejiang(Moderate)
	South(Smaller)

Analysis of Per Capita Price Differences:

1. Shanghai vs. Northern China:

The per capita prices show minimal variance between Shanghai and Northern China.

2. Shanghai vs. Other Chinese Regions:

There are significant differences in per capita prices between Shanghai and the southwestern, central, and southern regions of China, as well as the adjacent Jiangsu Province and Zhejiang Province. Particularly noteworthy is the substantial difference with the Central region, a moderate difference with the Southwest and Jiangsu-Zhejiang regions, and a smaller difference with the South.

3. Comparison of Northern China, Southwestern China, and Other Regions: Noticeable disparities are evident in per capita prices. Notably, there is a substantial difference with the Central region, a moderate difference with the Jiangsu and Zhejiang, and a minimal difference with the South.

Thus, there are significant variations in per capita prices among different regions, and the extent of these differences varies across regions.

For the Diamond Level, the significance P-value of the F-test is 0.489, manifesting that the Diamond Level shows no substantial impact on the Total Score. However, for the Black Pearl Restaurant List times and the "annual star merchant" times, the significance P-values of the F-test are 0.000*** each, suggesting a significant impact on the Total Score. Additionally, the influences of Geography and Cuisine on the Total Score are also notable, with the significance P-values of the F-test. These results indicate a significant impact of Black Pearl Restaurant Guide times, “annual star merchant” times, Geography, and Cuisine on the Total Score. This points towards the potential application of a polynomial regression, necessitating further investigation.

5. Conclusion：

In conclusion, several mathematical and statistical methods are used in this report to analyze some of the rules and requirements for the selection of “Black Pearl Restaurants Guide” in 2024. The data used in this report is first-hand information collected from the official Black Pearl Restaurants Guide and questionnaire survey released by the Meituan app. In exploring the detailed rules of the Black Pearl Restaurant score proportion, which Meituan has not published, there is a strong case for utilizing linear regression analysis. This represents a substantial breakthrough for me. After conducting a preliminary analysis, it becomes evident that the Black Pearl Restaurants are generally high regarded by the public. However, the median score in taste, environment and service surpasses the average score, and combined with the data table compiled in Excel, there are also a small number of restaurants with a lower score compared to other restaurants, around 4.0. In further analysis, it is also noticeable that the per capita price difference varies greatly from region to region. This report examines the significance of certain variables, such as the number of awards and the relationship between gastronomy and

geography, through hypothesis testing. The Wilcoxon sign-rank test for two samples showed that there was a significant difference between the “Won the annual star merchant” times and the “Black Pearl Restaurant lists” awards. Another mathematical method, the chi-square test, has been applied to show a strong correlation between cuisine and geographical region, with each region having its own regionally unique cuisine, but a wide variety of cuisines in regions with many award-winning restaurants. In addition, the most important method in this paper, linear regression analysis, generated a prediction model for the total score of Black Pearl Restaurant, and concluded that taste, environment, and service scores had a significant impact. Although this score scale has not been officially announced, this paper also combines ridge regression and questionnaire samples to confirm the model of this score scale. The paper also illustrates that the number of awards, including “Black Pearl Restaurant lists” times and, the “Won annual star merchant” times, regional location and cuisine also play a part in the score. This report underscores the importance of taste in opening a good “Black Pearl Restaurant” while also cautioning against underestimating other contributing factors.

This report marks the first academic report for the 2024 Black Pearl Restaurant Guide study. Although it is undeniable that taste has always been acknowledged as a crucial factor in evaluating a restaurant, Black Pearl extends beyond mere taste., This article delve into the scoring formula to explore multiple dimensions. Food lovers can evaluate a restaurant with a more critical eye after reading this article, while restaurant owners can learn from some of the conclusions of this report to better manage the restaurant and review and improve it in a timely manner. In terms of overall evaluation, the structure of this paper is relatively clear with coherent ideas. The mathematical methods used are clear and rigorous, incorporating some novel thinking and understanding. Additionally, the data sources are all first-hand data, which is easy to understand. Although the subject matter revolves around gastronomy, the whole report is presented in a mathematical style. The limitation of this report is that it does not fit a suitable model to determine whether a restaurant is a one-, two-, or three-diamond restaurant. This challenge arises due to the varied factors such as the existing region, Black Pearl Diamond grade, cuisine, price per capita, number of collections, number of reviews, opening date, “Won the annual star merchant” times, “Black Pearl Restaurant Guide times”, total rating, taste rating, environmental rating, environmental score. These factors prove insufficient for accurate judgment. In fact, there are traces in the official list. As a “once-in-a-lifetime” restaurant, there are only seven three-diamond restaurants in Chinese mainland, which is a minimal amount of data that is difficult to quantify from a mathematical point of view. In other words, these judging rules are also trade secrets of Meituan and Black Pearl Restaurant Guide, as well as some subjective elements of the Black Pearl expert jury. However, it is regrettable that the mathematical methods employed in this report are not particularly intricate.

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